

Andre Maciel

andremacielcasillas@gmail.com | 831-444-1838 | linkedin.com/in/andre-maciels | github.com/anmaciel1

EDUCATION

University of California, Santa Cruz | Santa Cruz, California

Expected June 2027

Bachelor of Science in Computer Science

Honors: Dean's List

EXPERIENCE

Software Engineer Intern - Salinas, California

Digital Nest

September 2025 - Present

- Built and deployed multiple production full-stack applications using **Node.js**, **Express**, **React**, and **Docker**, improving reliability and consistency across development environments.
- Built an end-to-end **Retrieval-Augmented Generation (RAG)** system that ingests user input, performs vector similarity search over indexed data, and generates **grounded LLM responses**, reducing hallucinated outputs in user-facing workflows.
- Designed and implemented **REST APIs consumed by React frontends**, enabling clean separation between client and server and improving data flow reliability across application features.

AI Research Associate - Santa Cruz, California

University of California, Santa Cruz

January 2026 - Present

- Conducted quantitative analysis on 120 students to evaluate the correlation between unprotected homework performance and proctored exam mastery in GenAI era.
- Built a full **Python** data pipeline (**Pandas**, **Seaborn**, **Matplotlib**, **Scikit-learn**) for **normalization**, **feature engineering**, **visualization**, and statistical validation.
- Identified statistically significant clusters of high homework (**>90%**) and low exam (**<70%**) performance.
- Detected right-skewed performance gap distributions and identified significant high-effort / low-mastery clusters affecting **51.6%** of students.

Engineering Intern - Monterey, California

Fab Lab

June 2023 - January 2026

- Led robotics instruction for 150+ elementary students across multiple cohorts, teaching programming fundamentals via Scratch and hands-on robotic design to cultivate early engineering interest.
- Designed and laser-cut custom project components, enabling students to apply fabrication and computational skills in integrated robotics projects.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, HTML, CSS, SQL

Frameworks/Libraries: Node.js, Angular, React, TensorFlow, Pandas, NumPy, Matplotlib, Scikit-learn

Tools: Git, REST APIs, Bash, Docker

RELEVANT COURSEWORK

Data Structures and Algorithms, Computer Systems and C Programming, Object Oriented Programming in Python, Intro to Web Development, Full Stack Application, Introduction to Machine Learning

PROJECTS

Pesticide Exposure Analysis | *Python, PostgreSQL, Pandas*

February 2026 - Present

- Designing a **data pipeline** and machine learning framework to analyze pesticide usage and potential exposure patterns in Monterey County using California Department of Pesticide Regulation data stored in a **PostgreSQL** relational database.
- Developing a normalized **SQL** schema and **Python** data-cleaning pipeline to transform raw pesticide use reports into structured datasets suitable for statistical analysis and machine learning.

Private Information Retrieval Demo | *TypeScript, React, Vite, Tailwind, Supabase*

July 2025 - September 2025

- Designed an interactive web app that visually demonstrated **Private Information Retrieval** (PIR) protocols.
- Built a modular client-server architecture to illustrate real-world PIR flow between encrypted queries and database responses.